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A symbiotic view of the origin of life at hydrothermal impact crater-lakes

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Submarine hydrothermal vents are generally considered as the likely habitats for the origin and evolution of early life on Earth. The theory suffers from the 'concentration problem' of cosmic and terrestrial biomolecules because of the vastness of the Eoarchean global ocean. An attractive alternative site would be highly sequestered, small, hydrothermal crater-lakes that might have cradled life on early Earth. A new symbiotic model for the origin of life at hydrothermal crater-lakes is proposed here. Meteoritic impacts on the Eoarchean crust at the tail end of the Heavy Bombardment period might have played important roles in the origin of life. Impacts and collisions that created hydrothermal crater lakes on the Eoarchean crust inadvertently became the perfect crucibles for prebiotic chemistry with building blocks of life, which ultimately led to the first organisms by prebiotic synthesis. In this scenario, life arose through four hierarchical stages of increasing molecular complexity in multiple niches of crater basins. In the cosmic stage (≥4.6 Ga), the building blocks of life had their beginnings in the interstellar space during the explosion of a nearby star. Both comets and carbonaceous chondrites delivered building blocks of life and ice to early Earth, which were accumulated in hydrothermal impact crater-lakes. In the geologic stage (∼4 Ga), crater basins contained an assortment of cosmic and terrestrial organic compounds, powered by hydrothermal, solar, tidal, and chemical energies, which drove the prebiotic synthesis. At the water surface, self-assembled primitive lipid membranes floated as a thick oil slick. Archean Greenstone belts in Greenland, Australia, and South Africa possibly represent the relics of these Archean craters, where the oldest fossils of thermophilic life (∼3.5 Ga) have been detected. In the chemical stage, monomers such as nucleotides and amino acids were selected from random assemblies of the prebiotic soup; they were polymerized at pores of mineral surfaces with the coevolution of RNA and protein molecules to form the 'RNA/protein world'. Lipid membranes randomly encapsulated these RNA and protein molecules to initiate a molecular symbiosis in a 'RNA/protein/lipid world' that led to hierarchical emergence of several cell components: plasma membranes, ribosomes, coding RNA and proteins, DNA, and finally protocells with a primitive genetic code. In the biological stage, the emergence of the first cells capable of reproduction, heredity, variation, and Darwinian evolution is the key breakthrough in the origin of life. RNA virus and prions may represent the evolutionary relics of the RNA/protein world that survived as parasites for billions of years. Although the proposed endosymbiotic model is speculative it has intrinsic heuristic value. Future experiments on encapsulated RNA virus and prions have the potential to create a synthetic cell that may confirm a coherent narrative of this hierarchical evolutionary sequence.

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1. Introduction

The scientific quest for the origin of life on early Earth remains one of the oldest and most elusive problems of science. Nobody knows where and how life originated. In the past century, the debate on the origin of life has centered entirely on the chemical evolution of the first cells from organic molecules by a long succession of chemical processes. In recent times, with the exploration of space, the study of the origin of life has shifted to a broader perspective including planetary beginnings and exobiology. This paper addresses a new symbiotic view of life, highlighting the cosmogeologic context of the origin of life, the development of prebiotic chemistry in a hydrothermal crater-lake environment, and finally the symbiotic assembly of the first cells, which were capable of reproduction, heredity, variation, and Darwinian selection. The purpose of this article is to offer four new views in the origin-of-life theories: (1) a hierarchical model of the origin of life; (2) hydrothermal impact

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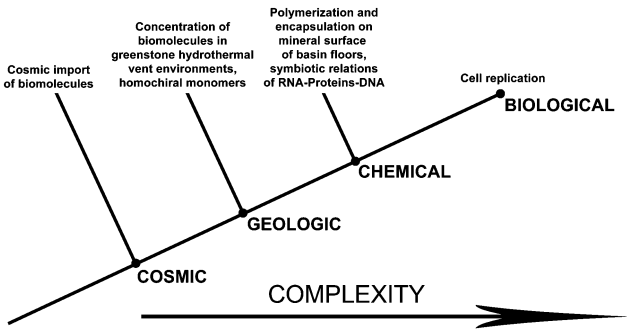

Fig. 1 Hierarchical origin of life. The origin of life can be viewed as four hierarchical stages with increasing complexity of biomolecules in the prebiotic world that led to the development of the first cells. These stages are: cosmic, geologic, chemical, and finally biological. All of these steps took place in the hot and dark environments of the hydrothermal crater basin, which may represent the Earth's oldest ecology (after Chatterjee[7]).

crater-lakes as a favorable milieu for the emergence of life; (3) the RNA world may have been the RNA/protein/lipid world; and (4) serial endosymbiosis of encapsulated RNA and protein molecules led to the origin of the first cells. I suggest an integrated and parsimonious model to address these questions that give the sequence and shape to the mystery of the origin of life.

The moment of the origin of life is considered to be a singular, non-linear chaotic event of historical contingency in time and space, an outcome of long chains of unpredictable antecedent states, one after the other.[1] However, the origin of life was not an isolated prebiotic chemical event. Using early Earth conditions at the dawn of the Archean as the starting point,[2–6] I argue that life probably arose in four, interconnected, hierarchical stages of increasing complexity—cosmic, geological, chemical, and biological[7] (Fig. 1). Each stage gave birth to a new sequence of biomolecules with new characteristics and increasing complexity that led to the emergence of the first cells. Each stage merged smoothly and continuously with the next stage. Such a multilayered hierarchy provided a kind of quality control at each assembly level.

Deep-sea hydrothermal vents, characterized by high temperature and high pressure, are generally considered as the primordial site—the crucible—for the emergence of life.[8–12] Like hot springs and geysers on land, hydrothermal vents form in volcanically active areas—often on mid-ocean ridges. Both types of submarine hydrothermal vents – acidic black smokers[2,8–10] and alkaline Lost City[11,12] – have been considered as possible cradles of early life. All these sites are geochemically reactive habitats, where microbial life thrives today around superheated water supporting diverse chemosynthetic ecosystems.[2] The submarine vent hypotheses have considerable appeal but have not been universally accepted, partly because many aspects of the proposed scenarios remain unconstrained from geological evidence of the early Earth environment and chemical constituents of living cells. In both regimes, the bonds that hold together vital cell components, such as nucleic acids, proteins, and membranes, face the danger of being torn apart as quickly as they are produced during the prebiotic stage to prevent further biogenesis. However, once the first cells similar to modern thermophiles appeared on the scene, they developed a high degree of thermostability.[2] This thermal paradox suggests that the crucible of life was cooler and quite different from the present day hyperthermal ecology of the black smokers and alkaline Lost City.

Both hydrothermal vent theories suffer from the 'concentration problem' of early life components because of the vastness of Eoarchean oceans. It's unlikely that life evolved in the submarine hydrothermal vents where the biomolecules would be greatly diluted and dispersed in open oceans, and no further reactions could occur. It's unlikely that life evolved in the open sea. To address this problem, I propose that a protective barrier of the potential crucible of life is needed for prebiotic synthesis. One crucial precondition for the origin of life is that comparatively simple biomolecules must have had the opportunities to form more complex molecules by segregation and concentration. So perhaps sequestered crater-lakes with hydrothermal systems were the best environments favoring organic reactions on early Earth. Life must have originated in some environment that was isolated from UV rays and global oceans to facilitate the concentration of prebiotic soup. Complex organic molecules are vulnerable to damage from sodium and chlorine in seawater.

Until recently, the only way to generate hydrothermal systems that we knew of was in a volcanic setting, but a recent study suggests that impacts also generate hydrothermal systems.[4–6] Here I propose that an impact-induced, hydrothermal crater-lake that hosted many microenvironments is a likely site for life's origin.[5] This hydrothermal crater lake, reconstructed from the early-Earth environment, is the terrestrial equivalent to the habitable environment of the Gale Crater of Mars, characterized by neutral pH, low salinity, and variable redox states of both iron and sulfur compounds.[13] Impact craters are of high interest in planetary explorations because they are viewed as possible sites for evidence of life. The high K^+/Na^+ ratio and relatively high concentrations of Zn, Mn, and phosphorous compounds of living cells support the geothermal terrestrial origin of life.[14] This finding challenges the widespread view that life originated in the sea,[10–12] but favors the freshwater impact crater basin as the likely crucible.[4–6,14,15] Hydrothermal systems in crater lakes are particularly regarded as sites where primitive life could evolve.[16]

The molecular structure and biochemical functions of living cells are an important starting point for addressing questions related to the origin of life. The minimum unit of life is the cell. Even the most rudimentary organism requires a certain minimum number of systems that perform different tasks to serve the function of a cell as a whole. These systems include: (1) the means to transmit heredity and reproduction (RNA and DNA); (2) a mechanism to obtain energy for metabolism and growth (proteins); (3) an enclosure to hold and protect these components from the environment (cell membrane); and finally (4) a unique principle to connect all these components together keeping the cell in operation (operating systems such as transduction). The importance of a cellular structure to keep these units in close enough communication to carry out successful reproduction has been clearly recognized in the origin of life debate.

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Much debate on the origin of life has centered on the question of whether genes or metabolism arose first. The defining characteristics of life include metabolism, and storage and duplication of genetic information. Protein is an essential component of metabolism, whereas nucleic acid is a prerequisite for replication. The most striking feature about a cellular life is the ubiquity and universality of this dual structure, replication and metabolism. I consider this dual function in a living cell to be the most compelling evidence for the existence of a 'RNA–protein world'. Perhaps from the beginning, there was a division of labor of cellular components—replication by RNA and metabolism by proteins, which are considered here as molecular fossils; one system does not try to become another. The duality of life, replication and metabolism, has an ancient origin before the birth of the first cells. The first cells required some mechanism by which a membrane could encapsulate these two functional units, RNA and protein molecules, for protection from the environment.

Among several competing hypotheses for how life arose on Earth, the 'RNA world' model is widely accepted[17–19] in spite of the complexity of RNA synthesis.[20–24] Although the RNA world theory is very attractive, no one has yet produced experimental proof of its viability. Naturally occurring RNA molecules possess very few of the specific enzymatic properties of proteins. Proteins make all RNAs and there is no evolutionary evidence that RNA was ever made by RNA. The synthesis and replication of RNA is dependent on the structural guidance provided by the enzyme. Here I propose a simpler origin of life, first outlined by Dyson[24] that RNA and protein molecules evolved synchronously in prebiotic environments to initiate an 'RNA/protein world'. Unlike Dyson's 'double-origin' model, I maintain that life arose on Earth only once, with the functions of replication and metabolism already present in rudimentary form and linked together from the beginning – and the planet has been continuously inhabited ever since. Perhaps, the function and feedback of replication and metabolism in the 'RNA/protein/lipid world' evolved gradually by molecular symbiosis inside protocells.[7]

2. Cosmic stage

The building blocks of life could have their beginnings in the tiny icy grains that make up the gas and dust in the interstellar space, and those icy grains could be the key to understanding how life arose on Earth.[25–32] These organic molecules were formed as a by-product of the explosion of a nearby star during the beginning of our Solar System (>4.6 Ga). Finding these biomolecules in an interstellar gas cloud means that important building blocks of life such as membrane-forming molecules, amino acids, nucleobases, sugars, and many organic compounds were seeded on the primordial Earth *via* meteorites and interstellar dusts.

The earliest history of the Earth during the Hadean period is barely recorded in the geologic record, but is reconstructed from the impact history of the Moon and inner planets. When the Earth formed some 4.6 Ga, the planet was sterile and inhospitable to living organisms, a seething caldron of erupting volcanoes, raining meteors, and hot noxious gases in a steam atmosphere. The Hadean Earth was a fiery globe of molten crust generated by relatively heavy bombardment of meteorite impacts; the atmosphere was thick and opaque to thermal radiation and the climate was dominated by internal heat.[25] As the frequency of impacts slowed down during the beginning of the Archean, Earth cooled, clouds formed, and the crust began to harden creating a terra firma. Understanding the environment within which the continental-type crustal rocks of Earth formed provides an important perspective and time frame on the nature of conditions of early Earth that encouraged life synthesis.

Meteorites played two distinct roles in the cosmic stage: exogenous delivery of key ingredients of life and endogenous synthesis at impact crater lakes for chemical evolution.[4–7,26] Some of the building blocks of life have been detected in Murchison meteorites such as nucleotides of DNA and RNA, 70 extraterrestrial amino acids, cell membranes, hydrocarbons, and carbonyl compounds.[26–32] As they are 4.6 Ga old, asteroids provide evidence that life's building blocks originated from icy interplanetary dust and were then carried to Earth, jump-starting life. Comets contain prodigious quantities of organic compounds including amino acids, adenine, ketones, quinones, carboxylic acids, hydrogen cyanide, polycyclic amino aromatic hydrocarbons, thioformaldehyde, acetaldehyde, sugars, cyanogens, cyanide, methanol, and ethanol.[26–31] Smaller comets or cometary dust probably delivered prebiotic molecules, which were freeze-dried and preassembled, to early Earth, but the hydrothermal crater basin was the crucible of life – the assembly factory where building blocks were linked randomly in a feedback loop until the final assembly of a first cell was produced by combinatorial chemistry and natural selection.[9,31] Recently, the Rosetta Mission of the European Space Agency (ESA) craft, Philae Lander already detected water and organic molecules on the surface of 67P/Churymov-Gerasimenko, a 4 km-wide comet that may shed new light on the composition of building blocks.[28]

NASA reproduces the interstellar building blocks of life in the laboratory. New experiments simulating conditions in deep space reveal that the complex building blocks of life could have been created on icy interplanetary dust and then carried to Earth, jump-starting life. Interstellar ice simulation experiments at the NASA Ames Research center in a cryogenic laboratory at 40 Kelvin have shown that UV radiation processing of presolar ices leads to more complex species. All the major building blocks of life were synthesized *via* irradiation of interstellar ice analogs (silicate minerals coated with simple molecules like H_2O, CO, CO_2, NH_3, and CH_3OH).[29,32] Many of these organic compounds formed in these experiments are also present in meteorites and cometary and asteroidal dusts, and some are relevant to the origin of life. These experiments suggest that building blocks of life may have begun in a rather unusual freezing environment of interstellar space during the planetary formation and were delivered to early Earth by meteorites as dust particles. Interstellar clouds or dust could serve as a shield for these organic compounds, absorbing much of the radiation on the outer edges and keeping it from reaching the interior. Most of the

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organic compounds found on early Earth had an interstellar heritage. Meteorites and interstellar dusts provide a record of the chemical processes that occurred before life began on Earth. In cosmic imperative, the earthlife is the dying ember of a star; it carries stardust in its DNA.

3. Geologic stage

The geologic stage encompassed the timing and environmental setting of the earliest life on Earth, which is critical for understanding the likely site for life's origin. Once the meteorite bombardment slowed down ~4 Ga, the Earth was still a one-plate planet before the inception of plate tectonics. Meteorites delivered interstellar ice that turned to water vapor from heat generated by their impacts, which cooled and condensed at Earth's surface to form liquid water. The liquid water began to accumulate on early Earth's surface to form global oceans. Ocean water in turn helped to dissolve excess CO_2 out of the atmosphere to make our planet habitable. Water vapor from the ocean surface began to condense and accumulate in the early atmosphere, and rained down with spectacular storms to begin the hydrosphere–atmosphere interactions. After relentless rain, the sky became blue and clear, and was rid of some of the toxic gases such as methane, hydrogen chloride, and ammonia of the Hadean atmosphere. Earth was no longer an alien, inhospitable world, but was transforming into a life-supporting environment.

The early crust was pockmarked with innumerable craters like the surface of the Moon, which were filled up with rain to form hydrothermal crater lakes. These crater basins offered a number of possible niches for prebiotic chemistry: impact generated hydrothermal vent systems; prolonged convective circulation of heated water and temperature gradient; sequestered sedimentary basins where cosmic ingredients could be mixed thoroughly with terrestrial hydrothermal chemicals to form more complex biomolecules, forming a complex solution called the prebiotic soup; and finally biopolymers, such as polynucleotides and polypeptides, could be concentrated in the nanopores of the mineral substrate for encapsulation.[4,5,12,17,32]

Not much land was exposed at the beginning of Archean time. Archean rocks (3.9–2.5 Ga) occur in relatively small areas as cratons in all continents. The continental crust existed in the Archean eon or even earlier (~4.35 Ga) as suggested by isotopic evidence from detrital zircons.[33] It is unclear how the earliest continental crust formed on Earth that was probably surfaced with basaltic material. Grieve *et al.*[34] calculated that large impact and collision events could lead to the formation of the felsic crust on the early Earth. The continental felsic crust begins to segregate from this mafic crust in an ocean island-like setting, where the impact-triggered upwelling mantle generates magma that crystallizes to form a new felsic crust. The numerous small felsic bodies of rock, termed protocontinents emerged in the Early Archean time, surrounded by a huge global ocean. These continental islands resembling the Seychelles or Sri Lanka would be the targets for meteoritic bombardments that would create terrestrial impact craters with hydrothermal systems (Fig. 2A). The Earth appears to have been inhabited as soon as sedimentary rocks appear in the geologic record at approximately 3.8 Ga.[3,8,35] Microfossils suggest that life emerged around 3.5 Ga or earlier in a narrow time window just after the end of the massive bombardment.[35–37]

The Early Archean impacts, instead of sterilizing the planet, enhanced organic synthesis by shock energy, allowed microbial life to emerge, and stimulated the biological community.[4–8,16,26] A global ocean, occasionally interrupted by protocontinents, made our Goldilocks planet habitable.[35] The early atmosphere was dominated by CO_2 and water vapor. Large impact events can generate a hydrothermal system if the craters are quickly filled up by rain water, and the heat sources come from shock-heated solid material, impact melt, and the central uplift. Many hydrothermal craters submerged in the depths of the oceans abounded. But thousands of circular, freshwater crater lakes appeared on the protocontinents.[34] Both meteorite impacts and torrential rains filled these craters with cosmic biomolecules, making them suitable geochemical environments for probiotic chemistry, particularly with respect to their hydrothermal systems.[4–8]

3.1 Global ocean

From the geologic record it appears that Earth was a watery planet in the early Archean time, periodically assaulted by impacts during the tail end of the Late Heavy Bombardment period.[25,26,35] Water is the matrix and medium of life. It is the ideal solvent for chemistries for life—dissolving many molecules, transporting them to reaction sites, while preserving their integrity. Water played a vital role in the prebiotic combinatorial chemical evolution, screening and selecting self-assembling macromolecules.[38] Water has many properties that are indispensable for the functioning of proteins and cells. Ice or water vapor will not do the same function. A bacterial cell is 70% water by weight, which serves as life's solvent. Ice is fairly common in the universe, found everywhere from vast interstellar clouds to several planets and moons. In contrast, liquid water is rare in the universe. Solid water cannot act as a lubricant for the molecular processes of life. Liquid water is essential for the kind of delicate chemistry that makes life possible.

There is active debate on the origin of water on early Earth, ranging from negligible to significant cometary contributions to terrestrial water. The deuterium-to-hydrogen ratio (D/H) in terrestrial ocean water, approximately 150 ppm, similar to the average of carbonaceous chondrites suggests that the principal sources of water might be chondritic asteroids, not comets.[28] Most likely, early Earth acquired water from the asteroid belt.

3.2 Impact origin of the Archean greenstone belts

The role of large meteorite impacts as triggers of major tectonic, thermal, magmatic, environmental, and mantle-crust processes during the Archean poses one of the fundamental questions of Earth science. The structure and environment of Archean Greenstone belts in Greenland, Australia, and South Africa (≥3.5 Ga) suggest that these basins may be the relics of the primordial impact craters, initially filled with impact-triggered melts of

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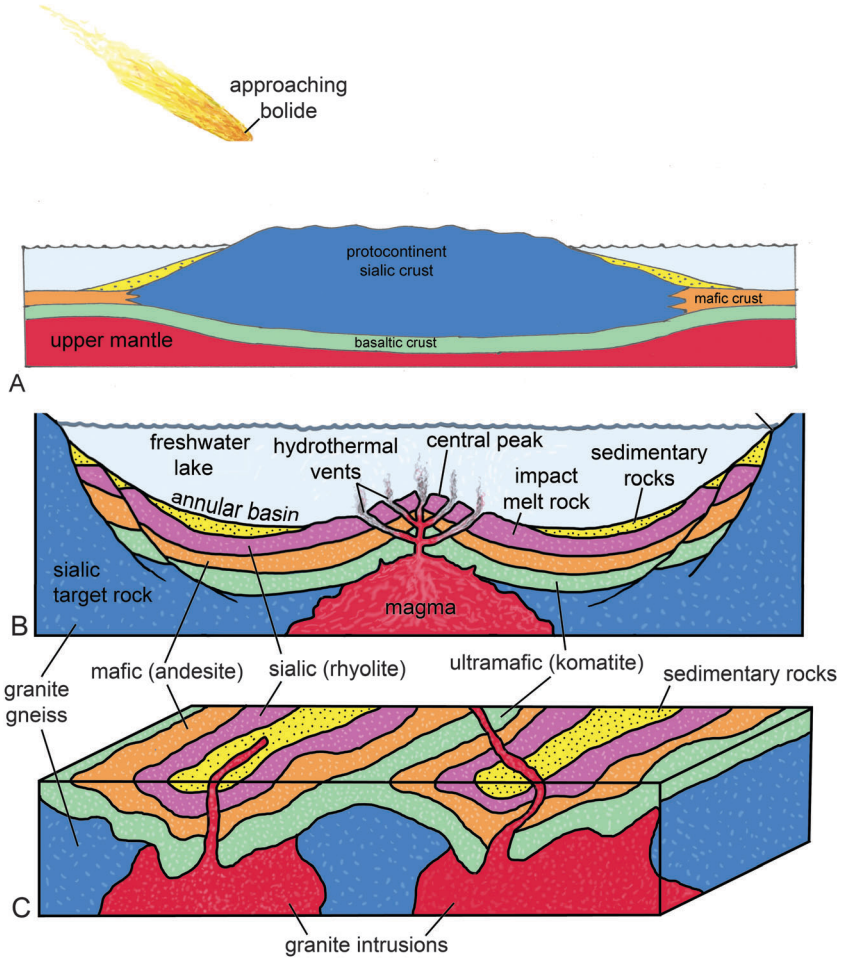

Fig. 2 Impact origin of Early Archean Greenstone belts. (A) An approaching bolide on a collision course to the protocontinental crust; (B) formation of a large, complex, freshwater crater lake with a hydrothermal vent system, which was sequestered by an elevated crater rim and separated from the ocean to become the cradle of life; the central peak of the crater was fractured to initiate the hydrothermal vent; the annual basin was the site of deposition of Algoma-type Banded Iron Formation and the chert; most of the oldest microfossils come from the chert deposits; (C) the crater was folded, metamorphosed, uplifted, and eroded to form the greenstone belt (after Chatterjee[7]).

ultramafic to mafic composition and later deformed and metamorphosed.[39,40] The Algoma-type Banded Iron Formation and volcanic chert in Greenstone belts suggest a hydrothermal-vent environment.[3] Four distinct spherule layers between 3.2 and 3.5 Ga from the Pilbara craton of Australia and the Kaapvaal craton of South Africa suggest that impact cratering processes on early Earth continued throughout the Paleoarchean era.[40–42] All these spherule layers represent the distal ejecta/fallout units from asteroid impacts in the Archean. The early bolide impacts that excavated large craters of the Greenstone basins of Greenland, Australia, and South Africa, deformed the target rocks, and created the suitable habitats for early life, such as thermophiles, anoxygenic photosynthetic bacteria, and cyanobacteria.[3,36,37] These Greenstone crater basins (~400 km across) developed hydrothermal systems at the central uplift[4] and were ideal habitats for early life, as reflected by the occurrence of the oldest microfossils as old as 3.5 Ga.[3,36,37] The Greenstone basins represent the oldest-known volcanic-sedimentary sequences on Earth and have been central in studies of early life. Archean Greenstone belts may be the terrestrial equivalent of the Gale Crater of Mars, which represents a large (150 km-diameter), pristine, freshwater, crater-lake deposit (3.8–3.5 Ga), and might be a potential site for the evidence of early Martian life.[13] Perhaps the pristine Archean Greenstone basin, before deformation and erosion, would mimic the Gale Crater of Mars.

The impact origin of a Greenstone belt by a large bolide is reconstructed here (Fig. 2). The young crust that formed on protocontinents was thin and thermally active, buoyant, and ductile during early Archean time.[5,25,34,39] A large, Eros-scale asteroid (~35 km across) slamming into such a crust could have excavated an enormous crater (~400 km diameter), about the size of these early Greenstone belts, which was quickly filled by cosmic water creating subsurface hydrothermal vent systems.[4,5,7,34,39–42] The architecture of the Greenstone basin mimics the morphology of a large complex crater.[43] In the cross-section, the central part of a Greenstone basin shows an anticlinal structure, which is flanked on either side by two synclinoriums, dominated by ultramafic, basic, and andesitic rocks.[44] Bilateral symmetry of the Greenstone

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basin resembles the geometry of a complex crater. Perhaps, the central part of the Greenstone basin represents the central peak of an immense crater, which was shattered across the central peak to initiate hydrothermal systems.[4–7,43] The annular basins, on either side of the central peak, contain typical Algoma-type Banded Iron Formations and the oldest microfossil-bearing chert, typical of hydrothermal vent environments.[3–7,36,37] The origins of Greenstone belts are controversial, often implicated in the early phases of plate tectonics.[44] However, a recent study suggests that plate tectonics did not start before ~3 Ga, about 500 million years later than the origin of the Greenstone basins.[45] It thus appears that some tectonic forces other than plate tectonics contributed to the origin of Greenstone belts. Impact cratering was still operative during the Eoarchean time,[40–42] and might be a viable alternative for the genesis of the Greenstone belts.

3.3 Small hydrothermal crater basins: crucibles of life

The earliest evidence for life on Earth comes from microfossils of simple bacteria in the Greenstone basins of Australia and South Africa that are about 3.5 billion years old. Ancient as their origins are, these bacteria (which are still around today) are already biologically complex. Life must have begun much earlier, perhaps as early as 3.8 Ga, as documented by the carbon isotope measurements of organic carbon in the Eoarchean Akilia and Isua supercrustal rocks of Greenland.[35]

Terrestrial hydrothermal impact crater-lakes have been proposed as the most plausible environments for the prebiotic synthesis of organic compounds necessary for life and may also have been sites for life's origin on Earth,[4–7] and, by analogy, on Mars.[16] They could have also provided a refuge for the earliest thermophilic life during late, giant-impact events. Complex impact craters (>2–4 km in diameter) are potentially capable of generating a hydrothermal system.[6] Evidence for impact-generated hydrothermal activity is recognized at over 70 of the ~180 craters on Earth, from the ~1.8 km diameter Lonar Lake structure, India, to the ~250 km diameter Sadbury structure, Canada. It is likely that hydrothermal crater lakes might have provided habitats for the origin and evolution of life on early Earth, and possibly other planets such as Mars.[6,16] Not only did hydrothermal crater lakes help sustain life by providing sources of heat, water, nutrients, and energy, but the shocked rocks in the crater basin offered shelter against harmful ultraviolet radiation at a time when Earth still lacked a protective ozone layer. Impact events generate shock pressures and temperatures that can melt substantial volumes of the target material. If the crater is filled with water to form a lake, the interaction of melt rocks with water in the near surface of the cater basin is capable of generating and sustaining a hydrothermal system.

It seems logical that the ideal crucibles of life would be small impact crater-lakes with hydrothermal vent systems to avoid the concentration problem of biomolecules, much smaller than the giant Greenstone crater basins. Small hydrothermal crater-lakes (~ 20 km diameter) similar to those of the Miocene Haughton crater of Canada and the Ries crater of Germany that host microbial communities would be the Archean analogues of the crucibles of life.[46] In general, the larger the crater, the longer is the duration of hydrothermal activity. The duration of hydrothermal activity for a smaller crater (~20 km) is poorly known, perhaps in tens of thousands of years. The Sudbury crater (~250 km diameter) probably retained hydrothermal activity at least for ~1 million years.[47] During the Eoarchean time, the crust was relatively thin and the heat flow was higher through the crust than at present, so these small craters retained hydrothermal activity for a longer period. The gradual cooling of the crater basin was advantageous for biogenesis, creating simple to complex organic compounds at different thermal gradients.

The building-block molecules such as amino acids, nucleotides, fatty acids, and sugars from cosmic sources began to accumulate in the crater basins, which were powered by the hydrothermal convection current, forming a complex solution of a thick, concentrated prebiotic soup. The soup was recharged with a host of chemicals from the hydrothermal vents that may aid in catalysis. Lipid cell membranes were first to form from cosmic ingredients by self-assembly.[29,31] The cosmic fatty bubbles extracted from the Murchison meteorite mimic cell membranes and show cell division in the laboratory. Most likely, primitive cell membranes floated on top of the water surface like a thick oil slick, but periodically they moved to the basin floor or to the lakeshore by convection current where they stuck to the mineral substrates for encapsulation of macromolecules such as RNAs and proteins.

The environmental complexity in a single setting was a necessary requirement for the origin of biologic complexity; and a wide range of environmental conditions must have been required to produce a living cell from organic precursors. A hydrothermal crater-lake creates different habitats, such as the central uplift, impact ejecta deposits, the annular basin, the crater rim region, and post-impact water and sediments in the lake, where a large number of chemical processes can occur simultaneously. The anoxic crater lakes in a pre-plate tectonic Earth are unique in geologic history and are very different from submarine black smokers, which we encounter today along the spreading ridge. The cold freshwater of the crater lake is heated by hot magma from the central peak and reemerges to form the vents that may reach high temperatures similar to the condition in Yellowstone Geysers, driving the convection of lake water with fluctuating temperatures at the water column.

Terrestrial, hydrothermal crater lakes possess all of the advantages of deep-sea hydrothermal vents that have been previously proposed for life synthesis such as prolonged circulation of heated water for mixing and concentration of prebiotic soup; various chemicals and ATP for chemosynthesis;[48] abundant catalytic surfaces of mineral substrates with nanopores and pockets for polymerization;[9,20] and nutrients for primitive life.[2,5,6,14,46] Moreover, they provide additional microenvironmental advantages in a single location including:

(1) Continental crater lakes cutting a variety of rock types provided a dynamic fluid mixing at the same time that favored prebiotic synthesis; this mix might be less available in a purely volcanic and tectonic venue.

(2) High crater rims for physical compartmentalization of cometary biomolecules for concentration.

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(3) Water basins with low ionic composition compatible with that of modern cells.[14]

(4) Diverse metal sulfides, clays, and zeolites as catalysts.[5,6,9,20]

(5) Impact-induced fracturing at the floor of the crater basin increased the surface area available for concentration of polynucleotides and polypeptides.[4,5]

(6) Wet and dry cycles of tidal pools near the surface of the crater favoring condensation reactions of polymers.[29,31]

(7) Moderate temperate ranges (~60 °C) and pH values near neutrality (pH 5–8) for stability of membrane vesicles.[31]

(8) Additional sources of energy including solar and tidal for evaporation and concentration of reactants.[4]

(9) Shocked rocks from impacts would offer shelter against harmful UV radiation when Earth still lacked a protective ozone layer.[6]

(10) Because of the huge amount of energy from impacts, impact melt rocks could keep the vent environments reactive for a long period of time during biosynthesis.[6]

(11) Hydrothermally altered and precipitated rocks can provide nutrients and habitats for life long after hydrothermal activity ceased.[6]

One advantage of multiple and interconnected microenvironments in a single setting of hydrothermal crater-lakes for biogenesis may be the close spatial proximity of different reactions to create more complex molecules in a feedback loop.[10]

4. Chemical stage

In the chemical stage, the dark, hot, and isolated environments of the crater basins served as the potential incubators of life.[4–8] The concentration of interacting molecules plays a fundamental role in the emergent complexity of the chemical system. Higher concentrations of biomolecules and higher temperatures tend to speed up chemical reactions. Energy flow in the chemical evolution is mediated by metabolic energy such as ATP.[48] Other sources of energy at the crater vents are reactive small molecules such as H_2S, CH_4, and NO_2 as well as thermal, solar, and tidal[11,15] (Fig. 3). Most of the chemical reactions in vent environments were endergonic with highly reactive free radicals far from the equilibrium stage. As the hydrothermal fluids react with the cosmic soup, their chemistry evolves. The convection current inside the craters mixed the hot, concentrated prebiotic soup thoroughly and caused simple chemicals to grow into larger, more complex ones by combinatorial chemistry with a chaotic mix of energy sources and organic compounds released from vents.[29,31] At this stage, Darwinian selection played an important role in sorting out about 5% of monomers from highly heterogeneous populations of monomers for molecular complementarity.[24]

4.1 Selection of homochiral molecules

All life today is homochiral and uses only L-amino acids and D-sugars. Chirality is pervasive in biological systems, but its molecular mechanism remains a puzzle. The selection of chiral molecules such as L-amino acid and D-sugar at the monomer level and a heterochiral relationship between these two specific isomers from a pool of isomers was fixed and selected in early history of biogenesis in building up proteins and nucleic acids. Homochiral monomers such as L-amino acids and D-sugar ribose were selected before they were polymerized. Why is the L/D mirror-symmetry broken by life? No one knows, but crystal faces of certain minerals such as calcite might have been involved.[20] All life today is homochiral and uses only L-amino acids and D-sugars because homochiral molecules fit into polymers. Most likely chiral purity was necessary for biological function and flow of information from DNA to RNA to proteins. At a later stage of biogenesis, a right-handed double helix of DNA would evolve, but the reason for this helical preference is not known.

4.2 Polymerization

The polymerization of monomers is the next key event in chemical evolution. Polymers are synthesized by condensation–dehydration reactions. The prebiotic monomers must be activated by loss of water molecules or by addition of phosphate for polymerization to be thermodynamically feasible.[29,31,49] Polymers of various kinds

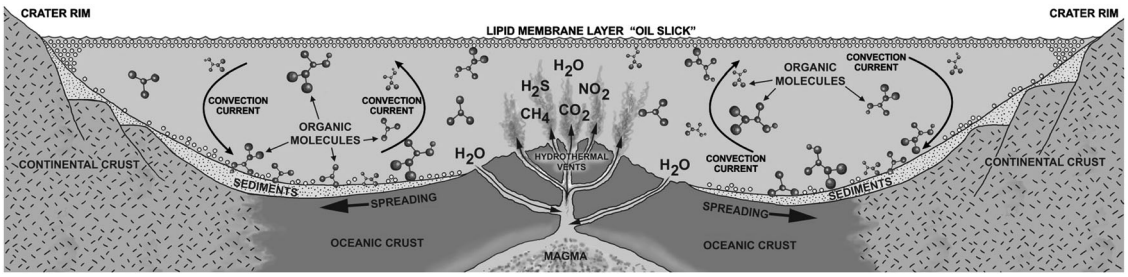

Fig. 3 The crucible of life. During the Early Archean period (~4 Ga), a small (~20 km diameter), freshwater crater basin with a hydrothermal vent system at the central peak would have offered a protected setting for the origin of life. The water in the basin was rich with organic molecules delivered by the comets and vents. At the water surface, primitive lipid membranes float as an oil slick. The mineral surface at the floor of the basin acted as a catalytic surface for the concentration and polymerization of monomers to form a 'RNA–protein world'. RNA and protein molecules began to interact and cooperate. The hot biotic soup was thoroughly mixed by convection current. Convection current also moved some of the lipid membranes to the basin floor, where they attach to the mineral surface for the encapsulation of RNA and protein molecules. Heat, gases, and chemical energy such as ATP released from the hydrothermal vent brewed the prebiotic soup that began to accumulate on the mineral substrate at the bottom of the basin (after Chatterjee[7]).

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were formed, joining in all sorts of assemblies, out of which a rare few turned out to contain the seeds of self-replication and catalytic activity. Infinite combinations must have taken place and infinite numbers must have failed and vanished. Certain monomers have the capacity to react with each other to form more complex polymers through a spontaneous self-assembly process in the presence of catalysts. Experiments have shown that these condensation reactions occur readily when growing polymers stick to small mineral particles.[9,20] Several mineral catalysts for polymerization such as clay and pyrites abounded in hydrothermal vent environments. These minerals provided the templates, the scaffolding, for the monomers such as nucleotides and amino acids to form polymers such as RNA and proteins. Molecular adhesion forces between microscopic layers of minerals brought these monomers close together and linked them to form polymers. Polymerization reactions were endergonic with high potential energy and lower entropy, which were mediated by chemical energy such as ATP, amino acids, sugars, and other organic compounds available in hydrothermal vent environments.[2,9,14,48] The Archean beaches around the crater basin at the interface of land, water, and air may be attractive sites for dehydration and polymerization of some monomers. Evaporation-concentration cycles in tidal pools on beaches of crater basins would have promoted dehydration reactions leading to polymerization.[29,31,49] Certain nitrogenous molecules from cosmic ingredients in the crater basin, such as cyanogen and cyanoacetylene, could act as condensation agents for dehydration of polymer chains.[30] In addition to the continued simmering of the primordial soup by convection current in hydrothermal vents, other mechanisms of concentration in pore spaces or mineral surfaces also increase the overall rates of chemical reactions. Once formed by polymerization, RNA and protein molecules began to concentrate in the nanopores and the stacking sequence of mineral substrates such as clays and pyrites respectively.[9,20] These biopolymers produced at mineral surfaces were likely small in size, simple in shape, and relatively random in sequence, creating a wide range of different chemical species and assortments from which few would be selected to produce the first cells.

4.3 The emergence of cell components

Life is associated with three classes of biomolecules—nucleic acids, proteins, and phospholipid membranes. These three subsystems seemed too different, their chemistries incompatible, and they arose from different precursors. The order of emergence of these three complex biomolecules, long before the first cells, is one of the most controversial topics in the origins of life story. Most likely, membranes appeared first because of their presence in cosmic ingredients.[29] Although phospholipids are universal components of cell membranes today, it seems improbable that such complex molecules were available on prebiotic Earth.[31] It is more likely that simple amphiphilic molecules such as lipids were precursors to phospholipid (or plasma) membranes. Meteorites such as carbonaceous chondrites may have brought this fatty lipid material to early Earth.[29,30] The hollow lipid spheres extracted from Murchison meteorites mimic cell membranes. They have enough mechanical strength to protect the biomolecules and to have given them some protections from environments. These simple, monolayer lipid molecules (micelles) began to concentrate on the water surface of the crater basin to form a thick scum[36] (Fig. 3). Micelles appear to be excellent candidates for prebiotic protocells because they have relatively high permeability to ions and small molecules permitting ion transports to maintain a neutral interior (pH 7) inside the membranes.[50] Most of the thermophiles have a single layer of the plasma membrane and cell wall.[2] A monolayer avoids the problem of delamination and shows a high degree of thermal stability. These lipid spheres have enough mechanical strength, both to have contained other cell components such as nucleic acids and proteins and to have given it some protection from outside. Once micelles encapsulate protein molecules in the chemical stage to initiate symbiosis, lipid membranes turn into plasma membranes to form rudimentary proton pumps.[7]

Next to lipid membranes, which came first in chronological order, RNA or proteins? This is a controversial issue. An RNA world that predated the modern world of proteins and DNA is a widely accepted model in the origin of life research.[17–19,51] The RNA world hypothesis proposes that self-replicating RNA molecules were precursors to all current life on Earth. The RNA molecules can act as information storage molecules due to their specificity in terms of interactions with other molecules. The RNA hypothesis has become dogma in scientific literature since the 1980s with the discovery of catalytic RNA such as ribozymes that nest at the heart of ribosomes—the protein-making machine, and if RNA makes proteins, surely RNA must have come first.[17–19,51] In the RNA world, an early form of life used no protein at all, but rather used RNA to play both genetic and catalytic roles. In this model, RNA enzymes (ribozymes) predate protein enzymes. Ribozymes performed a variety of catalytic functions in the RNA world, from metabolic biosynthesis to energy conservation. The discovery of ribozymes was a major event in support of the RNA world hypothesis. According to this hypothesis, RNA stored both genetic information and catalyzed chemical reactions in primitive cells. Only later in evolutionary time did DNA take over as the genetic material and proteins become the major catalyst and structural component of cells. RNA is regarded as the first self-replicating, naked molecule precursor to the first cell. Because of its multitasking functions, the RNA world hypothesis is widely and seductively popular in modern molecular biology.

However, the process by which the RNA world emerged remains a mystery, because no self-replicating RNA molecules exist today.[2–23] RNA is a very complicated molecule, each containing a sugar, phosphate, and one of four nitrogen containing bases as subunits. Nucleic acids form in living cells when nucleotides polymerize in the presence of enzymes. During chemical evolution, activated nucleotides probably polymerized on the surface of clay substrates to form primitive RNA molecules with a steady input of polypeptides. The RNA molecule is inherently fragile in the natural environment and constantly degrades into smaller fragments by hydrolysis, preventing faithful reproduction. Ribose in RNA is unstable and rapidly breaks down. No one has yet found intact RNA in fossils,

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ancient bone, or pollen grains. To perform its many tasks, RNA must be encapsulated.[29,31,49] This would imply, once more, that membranes arose much earlier than RNA. Nucleotides need enzymes to form RNA. Furthermore, RNA needs the help of protein molecules to undergo a kind of evolution. Similarly, primitive RNA needs polypeptide-like short chains of proteins to replicate. Primitive polypeptides must have provided some catalytic help for RNA to function.

The RNA world might have existed, but the exclusivity of RNA and the neglect of the protein and lipid membrane could have been overstated. Whether the primordial RNA first functioned in isolation, or in the presence of proteins and membranes, is still an open question. There is no evidence for biological takeover of ribozyme function by protein enzymes. Vent environments that could support RNA synthesis no doubt also spawned many other organic compounds—for example peptide and lipid-like molecules. Moreover, lipid membranes came from space almost intact.[29–32] It is irrational to think that vent environments exclusively created a load of nucleotides or RNA. Amino acids are easier to synthesize than RNA as the Miller-type experiment suggests. The versatility of RNA molecules does not prevent the formation of polypeptides in the vent environments, especially when polypeptides were the likely outcome in prebiotic synthesis.

Those who favor proteins as the first step argue that proteins were the first molecules that started life because of their underlying chemical simplicity and RNA molecules emerged later to support the replication of protein molecules.[9,20–23] There is no reason to think that noncoding proteins could not arise independently in vent environments. Of course, these proteins must be much shorter than the ones used in life today. Proteins are easier to make in the laboratory than RNA and are more durable and preserved in fossils. If we enlarge the definition of a protein to include any kind of polypeptide, then proteins may have preceded or coexisted with RNA, since amino acids were probably among the most abundant biogenetic building blocks available on prebiotic Earth and meteorites and they could have polymerized to make small proteins.[27,29,30] In addition, proteins are the most efficient catalysts known, but there is no obvious way for them to self-replicate. The protein-first theory faces a formidable challenge because of its lack of a self-replicating mechanism. Coding proteins could never be produced without RNAs. A metabolic life without replication is as absurd as a replicative life without metabolism. Variants of the protein-first hypothesis suggest that the protein world may have started as a world of thioesters.[21]

An alternative to the RNA world *via* a single catalytic polymer (ribozyme) is the possibility that both RNA and protein molecules were self-assembled in the vent environments and began to complement, communicate, and reciprocate with each other in a symbiotic relationship. The duality of replication and metabolism is the intrinsic property of life and must have appeared simultaneously before the origin of the first cells. Both RNAs and proteins worked in tandem to jumpstart the life assembly. The functions of living cells give important clues to these ancient interactions of RNA and proteins.

Here I advance an alternative hypothesis, the 'RNA/protein world' from the principle of parsimony that makes the fewest assumptions. In this scenario, the polypeptides fostered the chemical evolution of polynucleotides, and polynucleotides fostered the chemical evolution of polypeptides like a molecular symbiosis.[7,24] As a result, RNA and protein molecules emerged simultaneously on mineral substrates and coevolved by symbiosis. Such a prebiotic coevolution of RNA and noncoding protein molecules in vent environments offers a simpler view of the origin of life and fills the intractable gap between RNA and metabolism (Fig. 4). RNA did not work alone. Proteins made a major contribution to the mechanical support and replication of RNA molecules by its catalytic activity that sped up chemical reactions. In return, RNA molecules began to fold like proteins into different shapes and helped with rearranging the amino acids according to the specifications of their nucleotide bases to transfer information. This initial communication between cooperating RNA and protein molecules was the prelude to the origin of ribosomes and translation that would culminate once they were encapsulated. Only much later did the translation system of RNA-generated coding proteins come into play once the ribosome was installed by mutualistic symbiosis inside the membranes, as discussed later; the nucleotide sequences of RNA became linked to the amino acid sequences of proteins. The close linkage between the physical properties of amino acids, the genetic code, and protein folding was likely essential from the beginning, long before large, sophisticated molecules arrived on the scene. This symbiotic interaction of simple RNA and protein molecules was likely the key factor in the evolution from building blocks to organisms. During this chemical stage, primitive RNA and protein molecules worked together to solve many of the chemical problems associated with the emergence of life.

The coevolution of RNA and protein molecules in the vent environments in the RNA/protein world is corroborated recently from the phylogeny of ribosomes.[52,53] The ribosome is the large ribonucleoprotein that synthesizes all coded protein, and is assembled by the sequential binding of its constituent proteins to ribosomal RNA. The ribosome is the most ancient assembly in biology and has increased in size over billions of years by an accretion process that preserves the core. The ribosomal size is a proxy for complexity of organisms. Ribosomes are hybrid RNA–protein machines that use a nearly universal code to catalyze proteins. Phylogeny of the ribosome suggests that RNA and proteins coevolved and fused together to form the ribosomes.[52,53] The simplest ribosomes (those from bacteria and archaea) contain about 50 different proteins and three RNAs comprising about 45 nucleotides and two-thirds of the mass of ribosomes. No doubt the first ribosome had depended both on proteins and RNAs to function. This fusion of the ribosome could only form if RNA and proteins emerged simultaneously and were encapsulated by lipid membranes. Once they are enclosed, they began to interact and cooperate inside the protocells by serial endosymbiosis to form the ribosomes (Fig. 5). The phylogenetic origin of ribosomes shakes the foundation of the RNA world.[52,53]

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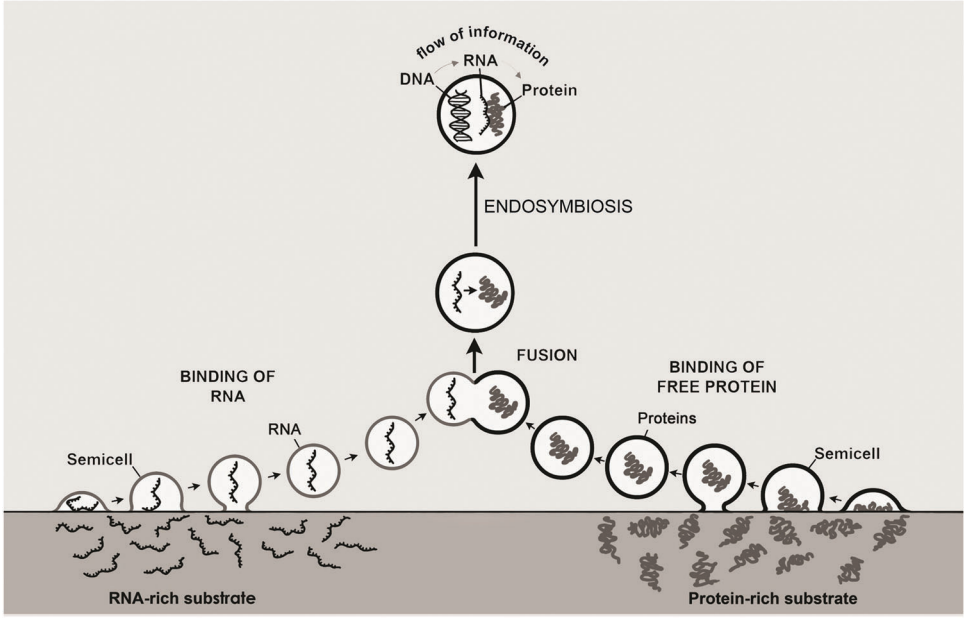

Fig. 4 Proposed model for the encapsulation of RNA and protein molecules by lipid membranes from clay and pyrite mineral substrates, respectively. Once these molecules were encapsulated, they began to fuse and interact; endosymbiosis led to the formation of DNA and the genetic code (after Chatterjee[7]).

4.4 Serial endosymbiosis

Margulis[54] championed the Serial Endosymbiosis Theory (SET) that the eukaryotic cell arose by the symbiotic merger of at least two kinds of bacteria—respiratory and cyanobacteria—inside a host cell to form their organelles such as mitochondrion and chloroplast, respectively. Here I propose a similar model of mutualistic endosymbiosis at a molecular level—beneficial associations among RNA, proteins, and lipid membranes, where lipid membranes were the host, and RNAs and proteins were prebiont molecules. At the floor of the crater basin, hollow lipid membranes, driven by the convection current, began to stick to the mineral substrates where they encapsulated RNA and protein molecules randomly along with the thick organic soup for nutrition and catalysis (Fig. 4). Once these RNA and protein molecules were enclosed, they began to interact and cooperate inside these lipid membranes to initiate symbiosis. Living cells consist of cooperating biomolecules. Thus, a reasonable way to envision the origin of life is to postulate the initial association of cooperating biomolecules inside lipid membranes. At this stage the 'RNA/protein world' turned into a highly creative 'RNA/protein/lipid world'. Symbiosis between RNA and protein molecules inside the cell membranes involving persistent physical contact and molecular coupling led to the multi-stage emergence of the first cells. Such a symbiotic coupling of genetics and metabolism may provide the most attractive origin scenario of all.

I identified 10 hypothetical stages for the emergence by life by serial endosymbiosis of encapsulated RNA and protein molecules (Fig. 5). These are:

(1) The encapsulation of RNA molecules from the clay substrate.

(2) The encapsulation of protein molecules from the pyrite substrate.

(3) First round of endosymbiosis between the protein and the lipid membrane to form a plasma membrane.

(4) The fusion of two protocells containing both RNA and protein molecules to initiate RNA–protein interaction; the 'RNA–protein world' became the 'RNA/protein/lipid world'.

(5) The second round of endosymbiosis and the origin of primitive ribosomes by fusion of RNA (a proto-RNA ribozyme) and protein molecules (a proto-protein enzyme). Once ribosomes were formed inside cell membranes, RNA molecules began to guide the specific proteins according to their nucleotide sequences in the form of a rudimentary translational system, while other RNA molecules acted as adaptors, each binding to a specific amino acid; in this stage, RNA began to synthesize proteins and proteins began to catalyze RNA; the co-dependence of RNA and proteins was built-in inside the cell membrane.

(6) The origin of coding RNA and proteins *via* ribosomes by a feedback loop; co-evolution of RNA and protein molecules began by autocatalysis, when genetically directed proteins appeared; the origin of translation linking the nucleotide sequences of genes to the amino acid sequences of proteins; prebiotic RNA and proteins from vents became redundant.

(7) The origin of retroviruses from RNA and the availability of reverse transcriptase; the protein that ushered the new, DNA storage format is a special enzyme called reverse transcriptase deployed by a retrovirus to create a DNA copy of its own genome.[55]

(8) The origin of uracil DNA (U-DNA) from RNA.

(9) The origin of thymine-DNA (T-DNA) from U-DNA with evolution of transcription. Most likely, DNA evolved from RNA in two steps:[55] the first step was most likely the formation of

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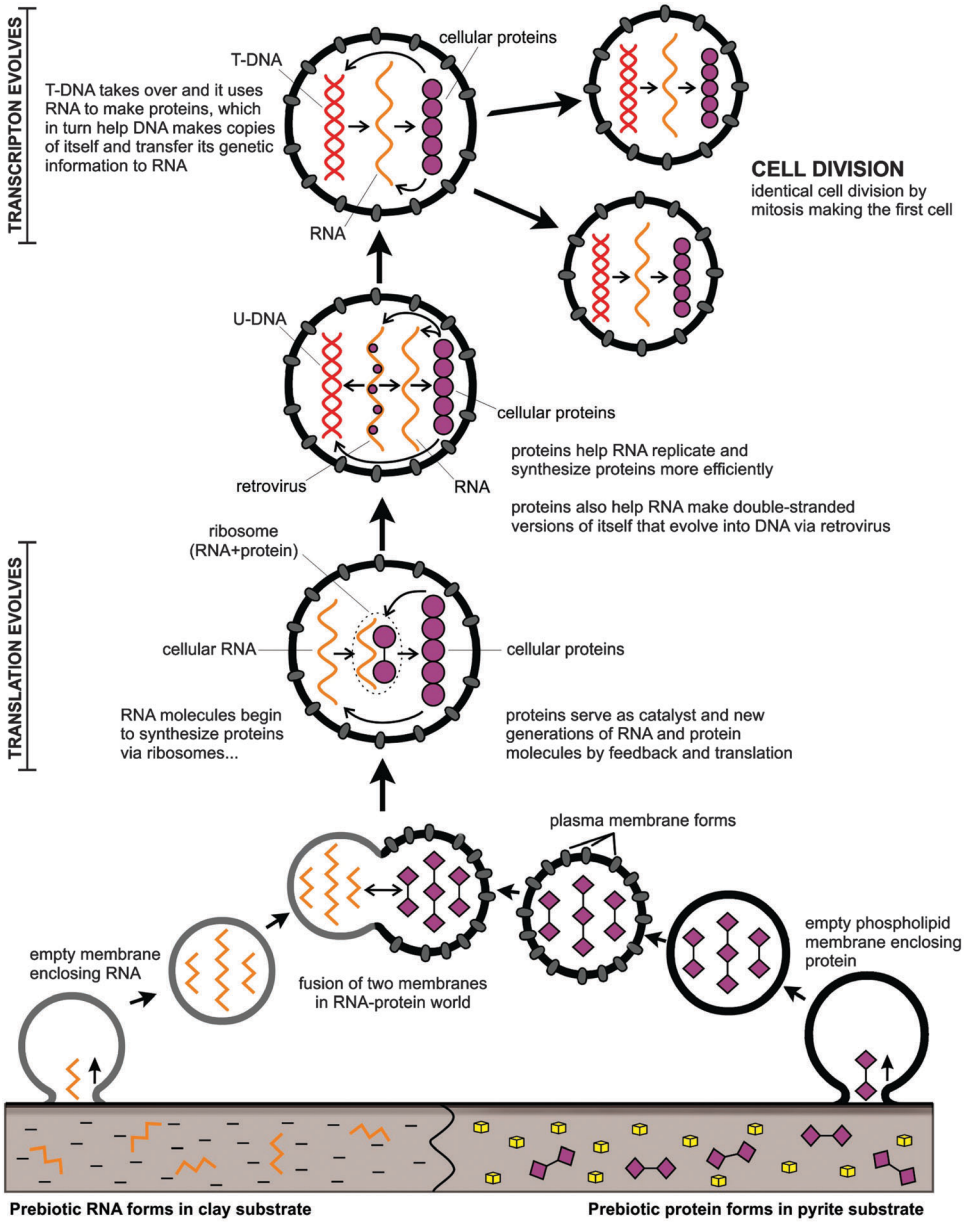

Fig. 5 Endosymbiotic theory of the origin of life in a 'RNA–protein–lipid world'. The proposed model for the emergence of the first cells by serial endosymbiosis of RNA and protein molecules inside lipid membranes showing 10 transitional stages. Once lipid membranes encapsulated RNA and protein molecules from clay and pyrite mineral substrates separately, they began to fuse; RNA and protein molecules began to cooperate and interact to produce ribosomes and new generations of RNA and coding protein molecules. Serial endosymbiosis led to the formation of DNA, genetic code, and finally the first cells, capable of reproduction, heredity, variation, and Darwinian evolution. At that tipping point, life had begun, and chemical evolution gave way to biological evolution (after Chatterjee[7]).

U-DNA after the reduction of ribose sugar to deoxyribose by chemically reactive free radicals at vents. The second step would be conversion of uracil into thymine by methylation to form T-DNA. The aqueous environment within the protocells plays a crucial role in the regulation of the supramolecular structure of DNA. DNA could be a final result of long term prebiotic natural selection. The primary molecular structure of DNA was selected because such structure enabled the further precise self-organization in the aqueous environment.[38] Once DNA molecules came on to scene, they became the primary genetic molecules, subduing RNA, which became an intermediary between DNA and proteins; genetic information began to flow from DNA to RNA to proteins.

(10) The reproduction of the first cell by binary fission.

The serial endosymbiotic theory also explains the hierarchical emergence of three basic information transduction systems: translation (RNA to protein), followed by transcription (DNA to RNA), and finally replication (DNA to DNA) (Fig. 5).

Two sets of properties distinguish living from nonliving systems: autopoiesis[56] and reproduction. Autopoiesis (self-making) refers to

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entities that are separated from an environment by an interface or a membrane and that metabolize (*i.e.*, chemically maintain and perpetuate their identity in fluctuating environments). Autopoiesis is considered a prerequisite for reproduction. In my model, stages 2–9 represent different stages of autopoietic entities that were capable of Darwinian evolution before the emergence of the first cells.

5. Biological stage: cell division

The biological stage represents the origin of cell division by binary fission, the asexual reproduction of a cell into two identical daughter cells. A cell needs to grow to twice its starting size and split in two. In bacterial replication, the circular chromosome is attached to the plasma membrane at about the midpoint of the cell. Before binary fission occurs, the cell must replicate its chromosome and segregate these copies to opposite ends of the cell. A contractile protein ring called FtsZ (the Z-ring) orchestrates chromosome separation and cell division in bacteria.[50,57] The origin of binary fission in the first cells remains elusive. Most likely, the first cell had a minimalist design – a small, spherical, unicellular organism with a bilayer phospholipid membrane and a ring-shaped coil of self-replicating DNA floating freely within the cell. It might have acquired a rudimentary form of cell division using physical mechanisms alone without the Z-ring or cell wall. The first cell most likely had achieved autotrophic survival-like thermophiles, absorbing chemical nutrients from the environment. Laboratory simulations hint at the solution of the primitive cell division mechanism. Lipid vesicles extracted from Murchison meteorites show primitive cell divisions spontaneously in the laboratory without the action of external forces.[29,31,49] When a mixture of these cosmic vesicles with amino acids and nucleic acids is shaken, these vesicles can trap organic molecules inside them and begin to interact. Apparently, these vesicles can take in substances from outside, through the lipid walls, and use them to build new walls and new contents. A large vesicle mimics a primitive kind of cell division. In a laboratory simulation, a genome-rich vesicle increases its size at the expense of an empty vesicle. When its size creates too much osmotic stress, pearling instability develops; the cell is stretched and divides into two, each keeping some of the original genome contents.[50,57] Perhaps, the initial cell divisions may have been accidental bursts, as cell sizes became increasingly large by accommodating more and more complex biopolymers. Random mutations in the copying process, and competition among other cells would then propel Darwinian evolution with variation, enabling these cells to divide precisely, adapt to their environment, and to compete with one another.

Once life evolved, deep and dark impact craters provided protected sedimentary basins for a hospitable environment for sustaining communities of primitive microorganisms like modern thermophiles[2] that began to spread globally. Direct evidence of very early cellular life ($\sim$3.5–3.2 Ga) comes from the Warrawoona Group in the Pilbara craton in Australia and the Swaziland Supergroup in the Kaapvaal craton in South Africa, where both thermophilic and primitive (anoxygenic) photosynthetic bacterial microfossils have been discovered.[36,37,58,59] The Archean Greenstone belts of Greenland, South Africa, and Australia represent the remnants of these ancient habitats of these early organisms such as thermophiles and cyanobacteria. Perhaps, some early colonies of benthic thermophilic bacteria diversified, changed their metabolic strategy, and invaded the coastal water; they mutated and evolved, and harnessed a near-infrared rather than visible sunlight to become anoxygenic photosynthetic bacteria. Finally these primitive photosynthetic bacteria evolved into cyanobacteria to make the early Earth oxygenated for the first time. A billion years later, primitive organisms began to split into three distinct domains – bacteria, archaea, and eukaryota – that make up the tree of life.[60]

6. Discussion

The geological setting for the proposed model is a hydrothermal crater-lake on a small continental island, surrounded by a global ocean $\sim$4 Ga. In this hydrothermal crater-lake, all of the cosmic building blocks of life could be assembled in this environment. There are four novel ideas in this view of life. First, I suggest that life arose through four hierarchical stages – cosmic, geological, chemical, and biological – as a series of ascending levels with increasing complexity of biomolecules, each stage leading to the next stage.

The second novel aspect of this proposal is that a small hydrothermal crater-lake ($\sim$20 km diameter, similar to the size of the Haughton crater in Canada and the Ries Crater in Germany) was a favorable milieu for the emergence of life during the Eoarchean time ($\sim$3.8 Ga). Today both Haughton and Ries craters host microbial communities and are modern analogues for the proposed crucible of life.

Third, I suggest that in the hydrothermal crater basins, RNA and proteins arose simultaneously at the mineral substrates, which were encapsulated randomly to start the RNA/protein/lipid world. The coevolution of RNA and protein molecules offers a well-integrated scenario for the emergence, role, and macromolecular components of living cells.[44,45] The collaboration of RNA and proteins was likely necessary for the spontaneous generation of complexity. I am not arguing against the importance of RNA in prebiotic chemistry, but I am suggesting that other biopolymers such as lipid membranes and proteins were critical partners during biogenesis.

The fourth new idea in this model is the serial endosymbiosis of the encapsulated RNA and proteins that ultimately gave rise to the first cells. These early protocells containing different species of RNA and proteins eventually fused providing a symbiotic stage in the origin of self-propagating life. The serial endosymbiosis led to hierarchical emergence of cell components including plasma membranes, ribosomes, customized RNA and proteins, and finally DNA with a rudimentary genetic code, followed by reproduction of cells to jump-start life. I suggest that the biomolecules of life—RNA and DNA, proteins, and phospholipid

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membranes—are the ultimate products of symbiosis. This hypothesis provides a portrait of a symbiotic evolution of life at biopolymer levels.

Based on evolutionary data, Woese[60] concluded that cellular life was preceded by a time of 'supramolecular aggregates', which represented a time when functional polymers worked together to solve many of the chemical problems associated with the emergence of life. The coevolution of RNA and protein molecules in the 'RNA/protein world' represents this stage. Woese called the transition between this time in chemical evolution and the start of cellular life The Darwinian Threshold. Woese[60] also predicted that translation (including ribosome) is older than RNA transcription, which is likewise older than DNA replication. In my model (Fig. 5), stage 1 represents the Darwinian Threshold, but stages 2–10 denote the first phase of Darwinian evolution. My model is consistent with Woese's chronology with the proposal that the crater vent environment was responsible for biopolymer replication in the RNA/protein world until the time that protein-based polymerases were synthesized by the ribosome in the RNA/protein/lipid world.

The general scheme of the hierarchical model of the origin of life proposed here is as follows: origin of building blocks of life in interstellar space (>4.6 Ga) → delivery of cosmic building blocks of life *via* meteorites to early Earth (~ 4 Ga) → mixing and concentration of cosmic organic compounds in sequestered hydrothermal crater basins supplied with local chemicals and various energy sources → self-assembly and concentration of lipid membranes → selection of chiral monomers such as left-handed amino acids and right-handed sugars → polymerization of monomers on mineral substrates → simultaneous appearance and cooperation of simple RNA and protein molecules on mineral substrates (RNA/protein world) → encapsulation of RNA and protein molecules to initiate serial endosymbiosis (RNA/protein/lipid world) → appearance of plasma membranes → origin of ribosomes → coding RNA and proteins with crude translation → right-helical DNA with transcription → first cells (thermophiles) capable of self-replication, heredity, variation, and Darwinian selection (≥ 3.5 Ga) → evolution of anoxygenic photosynthetic bacteria (~ 3.4 Ga) → evolution of cyanobacteria (~ 3.2 Ga).

7. Designing the protocell

The proposed symbiotic model of the origin of life is speculative but has intrinsic heuristic value. Future experiments on encapsulated RNA and protein molecules have the potential to create a simple synthetic cell that may confirm a coherent narrative of the RNA/protein world and symbiosis in cell evolution. The encapsulation mechanism of RNA and protein molecules proposed here is analogous to the use of phospholipids as pharmaceutical excipients in drug delivery systems, where a hollow droplet of the phospholipid, called the liposome, is filled with a drug of interest. In the quest to build a simple living system, much recent interest has focused on encapsulating a genetic or metabolic system inside membrane vesicles.[29,31,49,50,61,62] Craig Venter and team[62] have pioneered in creating a synthetic cell Synthia controlled by a synthetic genome of the bacterium *Mycoplasma mycoides*. Its genome has been assembled not inside a parent cell, but in a computer. Using the tools of synthetic biology, it might be possible to create evolving, replicating primitive cells enclosing simple RNA and protein molecules inside a cell membrane.

The strength of the 'RNA/protein/lipid world' hypothesis will ultimately depend on experiments that confirm symbiosis of simple RNA and protein molecules to produce analogs of pathways leading to synthetic cells. If we accept the dual origin of RNA and proteins in the hydrothermal vents of the crater basins, the next question is whether any modern analogs of these two molecules existing outside the living cells may be found. Modern RNA-viruses and protein-rich prions, though highly modified through billion years of parasitism, come close to the primordial RNA and protein molecules. Tracing the origins of these subcellular particles is difficult because they do not leave fossils. The central debating point of the origin of the RNA-viruses and prions is whether they are ancient, first appearing before the appearance of first cells, or evolved more recently after the development of first cells. Scientists have traditionally thought that viruses were relative latecomers to the evolutionary stage, emerging after the emergence of the first cells.[24] A contrary view from recent molecular phylogeny suggests that RNA-viruses, viroids, and prions are ancient subcellular particles that arose before the first cells.[63–65] Viral particles are by far the most abundant biological entities on our planet. Since viroids are naked loops of RNA, whereas viruses have a protein coat, viroids come closer to the early stage of polynucleotide evolution at mineral substrates of the crater basins.[64] Viroid genomes are extremely small in size, and they do not code proteins, as expected in the vent environments. Similarly, prions are small protein chains present in plants and animals that may be precursors to early noncoding polypeptides at the mineral substrates.[63] RNA virus and prions function exactly as the relics of ancient polynucleotides and polypeptides of the RNA/protein world in the vent environment, enduring in their primitive state for billions of years. As coding RNA and protein molecules coevolved inside the protocells *via* ribosomes (Fig. 4), RNA and protein molecules emerging from the mineral substrates became redundant in life synthesis. Since then, they survived as parasites of an evolving cell, existing in twilight zones between living and nonliving. If future experiments with membrane-bound RNA virus and prions result in the creation of a synthetic cell, capable of replication and heredity, the experiments may reflect the plausible pathways for the emergence of life on early Earth, confirming the existence of an RNA/protein world.

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